

AI-Driven Individual Savings Account Onboarding

Agile BRD

Document Control

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Executive Summary

The purpose of this BRD is to define functional and non-functional requirements for an AI-driven customer onboarding solution for Individual Savings Accounts in Nigerian banks. The solution will support multiple channels: in-branch, chatbot, mobile app, internet banking, and self-service kiosks, ensuring compliance with Central Bank of Nigeria (CBN), Nigeria Deposit Insurance Corporation (NDIC), Anti-Money Laundering/Know Your Customer (AML/KYC) standards, and other regulatory mandates.

Business Background & Problem Statement

Nigerian banks face increasing pressure to:

- Onboard customers faster while maintaining strict regulatory compliance
- Reduce fraud, identity theft, and money laundering risks
- Improve customer experience across digital channels
- Maintain audit-ready records and traceability

Current onboarding processes are often:

- Manual and fragmented
- Digital with human intervention
- Prone to human error and fraud
- Slow, leading to customer drop-off
- Less account subscription to the required services e.g. SMS, ATM
- Less account subscription to the digital platform e.g. Mobile and Internet Banking.

An AI-driven onboarding solution is required to automate identity verification, risk assessment, and compliance checks while aligning with Nigerian regulatory frameworks.

Regulatory & Compliance Landscape

This solution must comply with regulations issued by:

- Central Bank of Nigeria (CBN)
- Nigeria Deposit Insurance Corporation (NDIC)
- National Identity Management Commission (NIMC)

- Nigerian Financial Intelligence Unit (NFIU)
- Nigeria Data Protection Act (NDPA) & NDPR
- Bank Verification Number (BVN) Regulations
- Anti-Money Laundering / Combating Financing of Terrorism (AML/CFT) Rules

Business Objectives

1. Enable AI-assisted customer onboarding for savings accounts
2. Support Tiered KYC (Tier 1–3) per CBN guidelines
3. Ensure real-time compliance checks
4. Auto-enrol customers into:
 - Mobile Banking
 - Internet Banking
 - ATM/Debit Card
 - Transaction & balance alert services
5. Reduce fraud, impersonation, and abandoned onboarding journeys

In-Scope

- Individual Savings Account onboarding
- In-branch & digital self-service onboarding
- AI-driven:
 - Identity verification
 - Risk profiling
 - Fraud detection
- Digital banking subscription
- Regulatory reporting & audit logging

Out-of-Scope

- Corporate accounts
- Loan products
- Cross-border accounts

Stakeholders

Role	Responsibility
Retail Banking	Product ownership
Compliance	Regulatory adherence
Risk Management	Fraud & AML oversight
Internal Audit	Control & audit trails
IT / Digital Banking	Platform implementation
Branch Operations	In-branch execution
Customers	End users

High-Level Onboarding Process Flow

Channel Entry

- Customer initiates onboarding via:
 - Branch (assisted)
 - Mobile App
 - Internet Banking (pre-login)
 - Chatbot medium
 - USSD / QR referral (optional)

Customer Data Capture

Mandatory Fields

- Full Name
- Date of Birth
- Gender
- Phone Number
- Residential Address

- NIN
- BVN

Identity & KYC Verification (AI-Driven)

AI Capabilities

- OCR document capture (ID cards)
-
- Facial recognition & liveness detection
- Name & DOB matching
- Fraud pattern detection

Accepted IDs

- NIN
- International Passport
- Voter's Card
- Driver's License

Tiered KYC Logic (CBN-Compliant)

Tier	Requirements	Transaction Limits
Tier 1	BVN/NIN + basic bio-data	Low
Tier 2	Govt ID + address	Medium
Tier 3	Full KYC + proof of address	High

AI auto-assigns KYC tier based on validated data.

Risk, AML & Fraud Checks

- Sanctions screening (local & global watchlists)
- Politically Exposed Person (PEP) screening
- AI risk scoring (low / medium / high)
- Automated escalation for compliance review

Account Creation

Once cleared:

- Core Banking System creates savings account
- Unique Account Number generated
- Account status:
 - Active
 - Restricted (if pending review)

Digital Banking Subscription (Mandatory)

Mobile Banking App

- Auto-registration using phone number
- OTP verification
- Username creation
- AI-guided first login

Internet Banking

- Linked to same customer profile
- Role: Retail Individual
- Password & MFA setup

ATM / Debit Card

- Card request auto-initiated
- Card type: Verve / Mastercard
- Delivery:
 - Branch pickup OR
 - Home delivery
- PIN generation via secure channel

Alert & Notification Services

- SMS alerts (mandatory)
- Email alerts (optional)
- Push notifications (mobile app)

- Transaction, balance & security alerts

Compliance, Risk & Audit Requirements

Compliance Rules

- Consent capture for data usage
- NDPA-compliant data storage
- Mandatory KYC before activation
- Transaction limits enforcement
- Regulatory reporting (CBN & NFIU)

Risk Controls

- AI fraud scoring thresholds

Audit & Logging

- Immutable audit logs
- Time-stamped onboarding actions
- User/device/channel traceability
- AI decision explainability logs

Non-Functional Requirements

Category	Requirement
Security	Encryption at rest & in transit
Performance	< 3 min onboarding (digital)
Availability	99.9% uptime
Scalability	Nationwide usage
Accessibility	Low-bandwidth support

Agile Delivery Structure

Epics

1. Customer Onboarding Experience
2. AI KYC & Fraud Engine
3. Digital Banking Enablement
4. Compliance & Audit Framework

Sample User Story

As a retail customer

I want to open a savings account digitally

So that I can start banking without visiting a branch

Acceptance Criteria

- BVN verified successfully
- Account created within 3 minutes
- Mobile banking activated
- Alerts enabled automatically
- ATM card automatically subscribed and issued
- Internet banking enrolment automatically

Success Metrics (KPIs)

- % Digital onboarding success rate
- Average onboarding time
- Fraud incidence rate
- KYC rejection rate
- Active digital banking adoption
- Regulatory audit findings (zero tolerance)

Risks & Mitigation

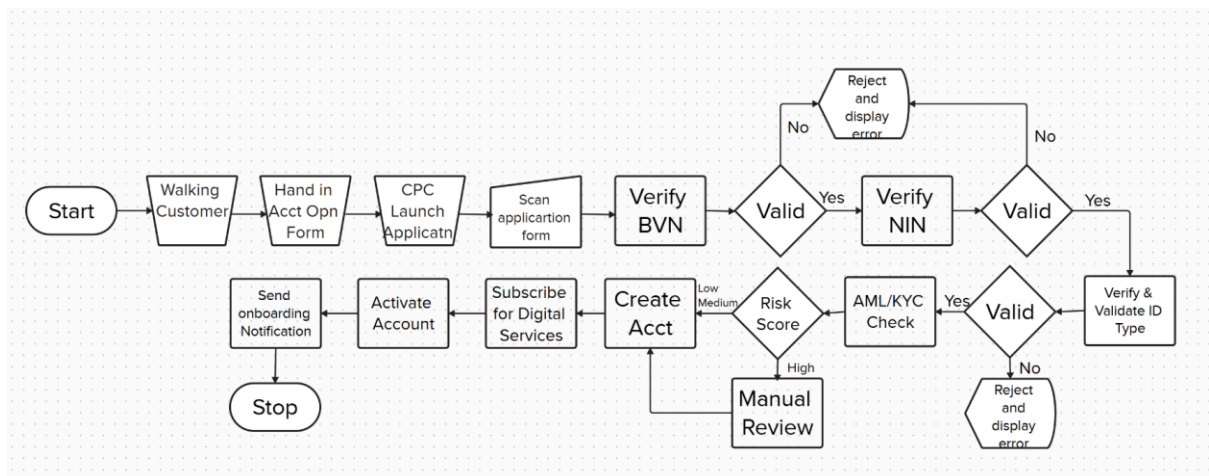
Risk	Mitigation
Identity fraud	AI + biometric validation

Risk	Mitigation
Regulatory breach	Automated compliance rules
Data privacy issues	Encryption & consent management
Poor digital adoption	Assisted digital onboarding

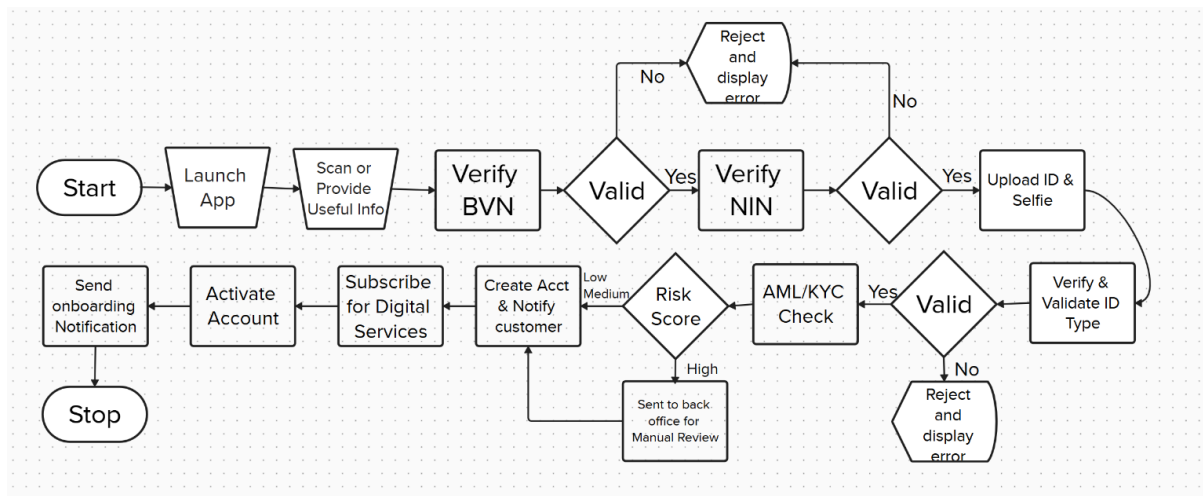
Open Assumptions

- BVN & NIN APIs are available
- AI models are explainable and auditable
- Bank has omnichannel core banking integration
- Digital banking subscription APIs are available
- ATM card subscription APIs are available
- ID verification APIs are available

In-Branch Onboarding BPMN



Digital/Chatbot Onboarding BPMN



Definition of Done (DoD)

- All acceptance criteria met
- Compliance sign-off obtained
- Audit trail verified
- Security testing completed
- AI model performance validated